PYTHON FOR DATA SCIENCE · LESSON 24 OF 29

APIs, JSON and Web Scraping with requests and BeautifulSoup

Collect data from the web. Understand HTTP, call REST APIs with requests (GET/POST, query parameters, headers, authentication, pagination, retries and rate limits), flatten nested JSON with json_normalize, and scrape HTML tables and pages responsibly with BeautifulSoup and pandas.read_html.

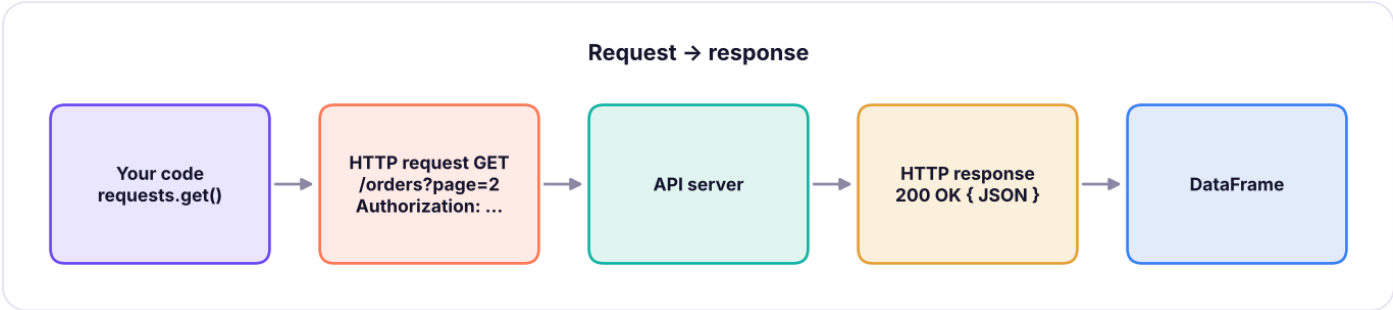
Python class 18 · ✦ requests 2.3x · httpx · BeautifulSoup 4 · Intermediate · ⏱ 100 min

By the end of this lesson you can

- Explain HTTP methods, status codes, headers and JSON responses
- Call REST APIs with parameters, headers and API keys stored safely
- Handle pagination, timeouts, retries and rate limits
- Flatten nested JSON into DataFrames with json_normalize
- Scrape HTML with BeautifulSoup and read_html, respecting robots.txt and terms of use

HTTP in five minutes

An **API** (Application Programming Interface) lets programs ask another system for data. Web APIs use **HTTP**: the client sends a **request** (method, URL, headers, optional body) and receives a **response** (status code, headers, body, usually JSON).



A REST API call.

Method	Meaning	Status code	Meaning
`GET`	Read	`200`	OK
`POST`	Create / send data	`201`	Created
`PUT` / `PATCH`	Replace / update	`400`	Bad request
`DELETE`	Remove	`401` / `403`	Not authenticated / not allowed
		`404`	Not found
		`429`	Too many requests (rate limit)
		`500` – `503`	Server error: retry later

A practice API

To keep this lesson runnable offline, we start a tiny local API that behaves like a real one (paginated orders, an API key, a rate limit). Every call below works the same way against real services.

Python

```
import json, threading, time
from http.server import BaseHTTPRequestHandler, HTTPServer
from urllib.parse import urlparse, parse_qs
ORDERS = [{"id": i, "customer": {"name": n, "city": c}, "items": [{"sku": f"P{i}", "qty": i
% 3 + 1, "price": 100.0 * i}], "status": s}
          for i, (n, c, s) in enumerate([("Asha", "Pune", "shipped"), ("Ravi", "Delhi",
"new"), ("Meera", "Kochi", "shipped"), ("Kabir", "Goa", "cancelled"), ("Zoya", "Pune",
"new")], 1)]
HITS = {"n": 0}
class Api(BaseHTTPRequestHandler):
    def log_message(self, *a): pass
    def send(self, code, body, ctype="application/json"):
        self.send_response(code); self.send_header("Content-Type", ctype);
self.end_headers(); self.wfile.write(body.encode())
    def do_GET(self):
        u = urlparse(self.path); q = parse_qs(u.query)
        if u.path == "/page": return self.send(200, "<html><body><h1>Prices</h1><table
id='p'><tr><th>Product</th><th>Price</th></tr><tr><td>Chair</td><td>₹4,500</td></tr><tr>
<td>Desk</td><td>₹12,000</td></tr></table><a href='/about'>About</a><p class='note'>Updated
daily</p></body></html>", "text/html; charset=utf-8")
        if self.headers.get("X-API-Key") != "demo-key": return self.send(401,
json.dumps({"error": "missing or bad API key"}))
        HITS["n"] += 1
        if u.path == "/flaky" and HITS["n"] % 2: return self.send(503, json.dumps({"error":
"busy"}))
        if u.path in ("/orders", "/flaky"):
            page, size = int(q.get("page", [1])[0]), int(q.get("size", [2])[0]); st =
q.get("status", [None])[0]
            rows = [o for o in ORDERS if not st or o["status"] == st]
            return self.send(200, json.dumps({"data": rows[(page - 1) * size: page * size],
"page": page, "next": page + 1 if page * size < len(rows) else None}))
        self.send(404, json.dumps({"error": "not found"}))
    def do_POST(self):
        body = json.loads(self.rfile.read(int(self.headers["Content-Length"])))
        self.send(201, json.dumps({"created": True, "echo": body}))
srv = HTTPServer(("127.0.0.1", 0), Api); BASE = f"http://127.0.0.1:{srv.server_port}"
threading.Thread(target=srv.serve_forever, daemon=True).start()
print("practice API running on this machine at", BASE.rsplit(":", 1)[0])
```

OUTPUT

```
practice API running on this machine at http://127.0.0.1
```

GET requests with requests

Python

```python
import requests, os
os.environ.setdefault("DEMO_API_KEY", "demo-key")                # in real life, set this outside the code
headers = {"X-API-Key": os.environ["DEMO_API_KEY"], "Accept": "application/json"}
r = requests.get(f"{BASE}/orders", params={"page": 1, "size": 2}, headers=headers, timeout=10)
print(r.status_code, r.url.replace(BASE, "BASE"), r.headers["Content-Type"])
payload = r.json()
print(payload["page"], payload["next"], payload["data"][0]["customer"])
bad = requests.get(f"{BASE}/orders", timeout=10)
print(bad.status_code, bad.json())
try: bad.raise_for_status()
except requests.HTTPError as e: print("HTTPError:", str(e)[:40])
```

OUTPUT

```
200 BASE/orders?page=1&size=2 application/json
1 2 {'name': 'Asha', 'city': 'Pune'}
401 {'error': 'missing or bad API key'}
HTTPError: 401 Client Error: Unauthorized for url:
```

Always set a `timeout`, check the status (`raise_for_status()`), and keep keys in environment variables.

POST requests

Python

```python
r = requests.post(f"{BASE}/orders", json={"customer": "Asha", "items": [{"sku": "P9", "qty": 1}]}, headers=headers, timeout=10)
print(r.status_code, r.json())
```

OUTPUT

```
201 {'created': True, 'echo': {'customer': 'Asha', 'items': [{'sku': 'P9', 'qty': 1}]}}
```

`json=` serialises the body and sets the `Content-Type` header for you. Use `data=` for form posts and `files=` for uploads.

Pagination, retries and rate limits

Python

```
def fetch_all(path, params=None, max_pages=50):
    rows, page = [], 1
    with requests.Session() as session:                          # re-uses the connection
        session.headers.update(headers)
        while page and page ⩽ max_pages:
            resp = session.get(f"{BASE}{path}", params={**(params or {}), "page": page,
"size": 2}, timeout=10)
            resp.raise_for_status()
            body = resp.json(); rows += body["data"]; page = body["next"]
    return rows
orders = fetch_all("/orders"); print(len(orders), "orders fetched across pages")

def get_with_retry(url, tries=4, backoff=0.2, **kw):
    for attempt in range(1, tries + 1):
        resp = requests.get(url, timeout=10, **kw)
        if resp.status_code not in (429, 500, 502, 503, 504): return resp
        wait = float(resp.headers.get("Retry-After", backoff * 2 ** (attempt - 1)))     #
exponential backoff
        print(f"  attempt {attempt}: {resp.status_code}, retrying in {wait:.1f}s");
time.sleep(wait)
    resp.raise_for_status()
print(get_with_retry(f"{BASE}/flaky", headers=headers).status_code)
```

OUTPUT

```
5 orders fetched across pages
  attempt 1: 503, retrying in 0.2s
200
```

Respect rate limits: read the API's documented limits, honour `Retry-After`, cache results, and sleep between calls.

Flattening nested JSON

Python

```
import pandas as pd
pd.set_option("display.width", 160)
flat = pd.json_normalize(orders, sep="_")
print(flat.drop(columns="items"))
lines = pd.json_normalize(orders, record_path="items", meta=["id", "status", ["customer",
"city"]], sep="_")
lines["value"] = lines.qty * lines.price
print(lines)
```

OUTPUT

```
   id     status customer_name customer_city
0   1    shipped          Asha          Pune
1   2        new          Ravi         Delhi
2   3    shipped         Meera         Kochi
3   4  cancelled         Kabir           Goa
4   5        new          Zoya          Pune
  sku  qty  price id     status customer_city   value
0  P1    2  100.0  1    shipped          Pune   200.0
1  P2    3  200.0  2        new         Delhi   600.0
2  P3    1  300.0  3    shipped         Kochi   300.0
3  P4    2  400.0  4  cancelled           Goa   800.0
4  P5    3  500.0  5        new          Pune  1500.0
```

`record_path` explodes a nested list into rows; `meta` copies parent fields onto each row.

Real public APIs to practise with

API	Data	Notes
data.gov.in	Indian government open data	Free API key
RBI DBIE, World Bank, IMF	Economic indicators	`wbgapi` package for the World Bank
Open-Meteo	Weather forecasts and history	No key
GitHub REST API	Repositories, issues	Token raises limits
Alpha Vantage, yfinance	Stock prices	Free tiers, rate-limited
OpenAI, Anthropic, Gemini	LLMs	Paid keys, see GenAI track

Python

```
# Example (needs internet): daily max temperature in Pune from Open-Meteo
r = requests.get("https://api.open-meteo.com/v1/forecast",
                 params={"latitude": 18.52, "longitude": 73.86, "daily":
"temperature_2m_max", "timezone": "Asia/Kolkata"}, timeout=10)
weather = pd.DataFrame(r.json()["daily"])
```

Web scraping

When there is no API, you can parse HTML. **BeautifulSoup** navigates the HTML tree; `pandas.read_html` grabs `<table>` elements directly.

Python

```
from bs4 import BeautifulSoup
from io import StringIO
html = requests.get(f"{BASE}/page", timeout=10).text
soup = BeautifulSoup(html, "html.parser")
print(soup.h1.text, "|", soup.find("p", class_="note").get_text(strip=True), "|",
[a["href"] for a in soup.find_all("a")])
rows = [[td.get_text(strip=True) for td in tr.find_all(["td", "th"])] for tr in
soup.select("table#p tr")]
print(rows)
t = pd.read_html(StringIO(html))[0]
t["Price"] = t["Price"].str.replace(r"[₹,]", "", regex=True).astype(int)
print(t, t.Price.sum())
srv.shutdown()
```

OUTPUT

```
Prices | Updated daily | ['/about']
[['Product', 'Price'], ['Chair', '₹4,500'], ['Desk', '₹12,000']]
  Product  Price
0   Chair   4500
1    Desk  12000 16500
```

BeautifulSoup	Finds
`soup.find("tag", class_="x")`	First match
`soup.find_all("a")`	All matches
`soup.select("div.card > h2")`	CSS selectors
`el.get_text(strip=True)` , `el["href"]`	Text and attributes

For pages built by JavaScript, use **Playwright** or Selenium to drive a real browser, or look in the browser's Network tab for the underlying JSON API.

WATCH OUT **Scrape responsibly and legally**

Check the site's **terms of use** and **robots.txt**, prefer official APIs, identify your scraper with a User-Agent, go slowly (one request every few seconds), cache pages, and never collect personal data without a lawful basis (India's DPDP Act 2023, GDPR). Logged-in or paywalled content is usually off-limits.

Summary

KEY TAKEAWAYS

- HTTP: methods (GET, POST…), status codes (200, 401, 404, 429, 5xx), headers and JSON bodies.
- `requests.get(url, params=, headers=, timeout=)` ; `r.json()` ; `r.raise_for_status()` ; keys from environment variables.
- Loop over pages with a `Session` ; retry 429/5xx with exponential backoff and honour `Retry-After` .
- `pd.json_normalize(data, record_path=, meta=)` flattens nested JSON.
- Scrape with BeautifulSoup or `pd.read_html` only when allowed; prefer APIs and be polite.

Quick quiz

6 questions · tap an answer to check it

1. What does status code 429 mean?

A Not found

B Too many requests (rate limit)

C Server error

D Success

Answer: B. Slow down and retry after the indicated time.

2. Why set `timeout=` on every request?

A It speeds up requests

B Without it, a request can hang forever

C It is required by APIs

D To retry automatically

Answer: B. Typical values are 5–30 seconds.

3. Where should an API key be kept?

A In the URL in your notebook

B In an environment variable or secrets store

C In a GitHub README

D In a comment

Answer: B. Read it with `os.getenv`.

4. What does `json_normalize(..., record_path="items")` do?

A Deletes the items field

B Creates one row per element of each record's items list

C Converts to CSV

D Sorts the data

Answer: B. `meta` keeps parent fields.

5. Which is the best first option to get data from a website?

A An official API or data download

B Scraping every page quickly

C Copy-paste

D A screenshot

Answer: A. Scraping is a fallback.

6. Which tool can read every `<table>` on a page into DataFrames?

A `requests.post`

B `pd.read_html`

C `json.loads`

D `soup.h1`

Answer: B. It returns a list of DataFrames.

EXERCISE **Practice**

(1) Using the Open-Meteo example (on your machine), fetch 7 days of forecasts for three cities and combine them into one DataFrame. (2) Write `fetch_all` for an API that uses `offset` / `limit` instead of pages. (3) Scrape the list of tables from a Wikipedia page of your choice with `read_html` and clean one of them.

Show solution (worked answer)

Python

```
cities = {"Pune": (18.52, 73.86), "Delhi": (28.61, 77.21), "Kochi": (9.93, 76.27)}
frames = []
for name, (lat, lon) in cities.items():
    r = requests.get("https://api.open-meteo.com/v1/forecast", params={"latitude":
lat, "longitude": lon, "daily": "temperature_2m_max", "timezone": "Asia/Kolkata"},
timeout=10)
    frames.append(pd.DataFrame(r.json()["daily"]).assign(city=name))
df = pd.concat(frames)
def fetch_offset(url, limit=100):
    out, offset = [], 0
    while True:
        batch = requests.get(url, params={"offset": offset, "limit": limit},
timeout=10).json()
        if not batch: return out
        out += batch; offset += limit
```

INTERVIEW CORNER **Questions you may be asked**

Q: How do you collect data from a paginated API reliably? Use a session, loop until there is no next page/cursor, set timeouts, retry 429/5xx with backoff, respect rate limits, log progress, and make the job restartable.

Q: API vs web scraping? APIs are stable, documented and permitted; scraping depends on page structure, can break, and has legal and ethical limits.

Q: How do you turn nested JSON into a table? `pd.json_normalize` with `record_path` and `meta`, or explode list columns and expand dict columns.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.